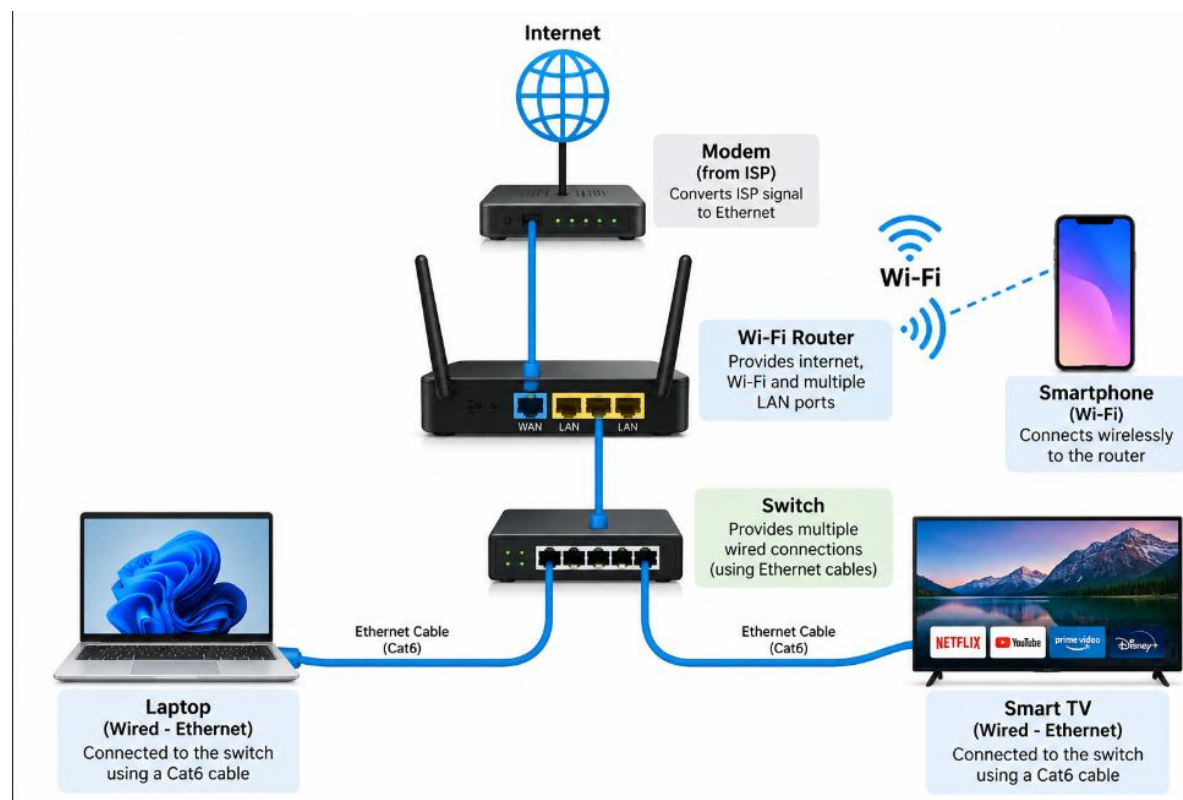


Network Devices

Task – 1 : Draw and label a simple home network diagram showing a router, switch, and three devices (like a laptop, smartphone, and smart TV), indicating how each device connects to the network.

Solution : Simple Home Network Diagram

A home network can contain a **router, switch, laptop, smartphone, and smart TV**. The router provides connectivity to the Internet, while a switch can connect multiple wired devices.



Task – 2 : Create a comparison table listing at least four network devices (router, switch, hub, access point) and describe one real-world use case for each, referencing apps or services you use daily (e.g., WiFi for streaming on YouTube, LAN gaming, etc.).

Solution : **Comparison of Network Devices**

Network Device	Main Function	Real-World Use Case	Example
Router	Connects different networks and forwards packets between them	Connects a home LAN to the Internet	Browsing Instagram or ordering from Zomato
Switch	Connects devices within a LAN and forwards Ethernet frames to the appropriate device	Connects computers in a gaming zone or computer lab	LAN gaming
Hub	Sends incoming data to all connected ports	Basic/legacy small network setups	Older laboratory networks
Access Point (AP)	Provides wireless network access to devices	Provides Wi-Fi to phones and laptops	Streaming YouTube over Wi-Fi

Task – 3 : Write a short scenario where you explain how data travels from your phone to the internet when you order food on Zomato, mentioning the role of each network device involved (router, switch, access point).

Solution : Ordering Food on Zomato

When I place a food order on Zomato, my phone sends the request through **Wi-Fi to the access point**. The access point forwards the traffic into the wired network, and the **router** sends the IP packets toward the Internet. If a separate **switch** is present in the network, it forwards Ethernet frames between the access point and router. The ISP then carries the traffic across the Internet to Zomato's servers. The response follows the reverse path and eventually reaches my phone.

Role of Each Device

- **Access Point:** Provides Wi-Fi connectivity to the smartphone.
- **Switch:** Connects wired network devices and forwards Ethernet frames within the LAN.
- **Router:** Connects the local network to other networks and forwards IP packets toward the Internet.

Task – 4 : Given this situation: you have a WiFi router, but your gaming PC only has an Ethernet port and is far from the router. Suggest two different network devices you could use to connect your PC to the internet and explain how each would work in this setup. Hint : Think about devices like switches, powerline adapters, or WiFi extenders.

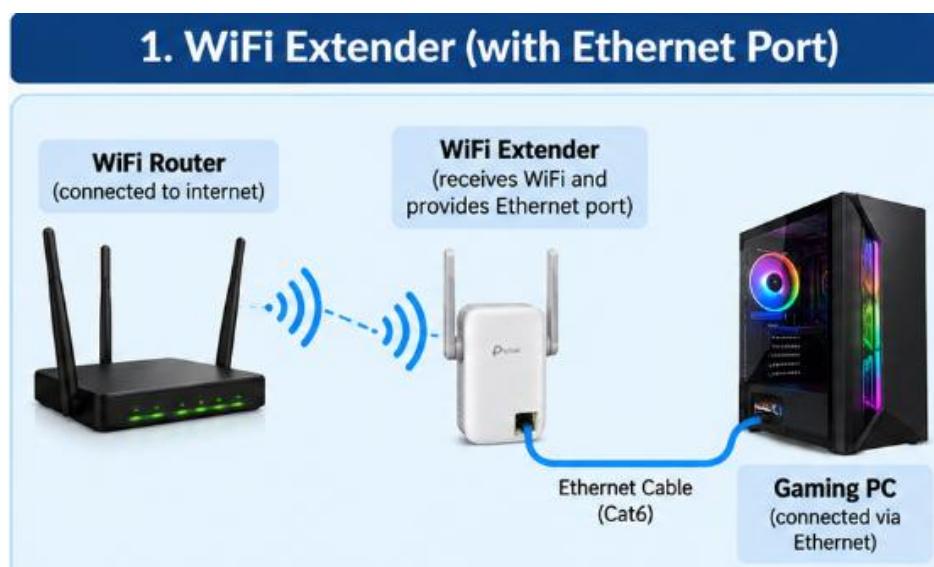
Solution : **Gaming PC Far From Wi-Fi Router**

My gaming PC only has an **Ethernet port**, but the Wi-Fi router is far away.

A **switch alone will not solve the distance problem** because the switch itself still needs a network connection to the router. Two suitable options are:

Option 1: Powerline Adapter

A **Powerline adapter** uses the building's electrical wiring to carry network traffic.



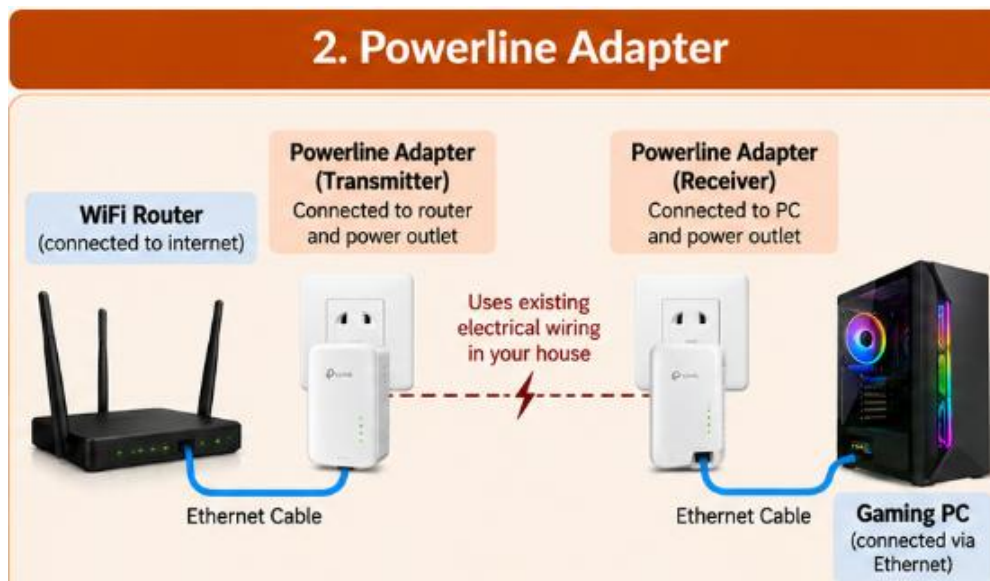
How it works

One powerline adapter connects to the router using Ethernet. The second adapter is placed near the gaming PC and connects to the PC using another Ethernet cable. Network data travels between the adapters through the electrical wiring.

Advantage: The PC gets a wired Ethernet connection without running a long Ethernet cable across the room.

Option 2: Wi-Fi Bridge / Wireless Ethernet Adapter

A **Wi-Fi bridge** or wireless Ethernet adapter can receive the router's Wi-Fi signal and provide an Ethernet connection to the gaming PC.



How it works

The wireless bridge connects to the router over Wi-Fi and then provides an Ethernet port for the gaming PC. The PC therefore does not need built-in Wi-Fi.

Advantage: No long Ethernet cable is required, and installation is relatively simple.